

Forecasting International Tourist Arrivals to Vietnam using SARIMA/ARIMAX

The Role of Online Search Behavior

Time Series Analysis — Individual Project

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Abstract

This study investigates whether Google Search Trends improve forecasting accuracy for international tourist arrivals to Vietnam (2010–2023) using an ARIMAX framework, and how this econometric approach compares to machine learning alternatives — Random Forest and XGBoost. We train all models on pre-COVID data (2010–2019) and evaluate on the test period (2020–2023), which includes the COVID-19 structural break and subsequent recovery.

Under static forecasting, machine learning models achieve the lowest RMSE (Random Forest: 8.43 on the log scale), outperforming ARIMAX (9.28) and SARIMA (9.40). However, under walk-forward rolling forecasting — which simulates real-world monthly model updating — ARIMAX achieves the best performance (RMSE = 2.24), reversing the static ranking.

The estimated coefficient on `log_search_trends` is $\hat{\beta} = 0.40$ ($p = 0.013$), suggesting a 1% increase in search activity is associated with a 0.40% increase in arrivals. Granger causality tests on pre-COVID data are not significant, indicating that Google Trends functions as a *coincident* indicator of demand collapse rather than a traditional leading indicator. We conclude that ARIMAX with search trends is the recommended specification for operational monthly forecasting, offering superior rolling accuracy, interpretable demand elasticities, and valid confidence intervals.

Keywords: ARIMAX, SARIMA, Tourism Demand Forecasting, Google Trends, COVID-19, Random Forest, XGBoost, Walk-forward Forecast

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1 Introduction

1.1 Background and Motivation

Tourism is one of Vietnam’s most significant economic sectors, contributing approximately 9.2% of GDP in 2019 and employing over 2.5 million people directly. International tourist arrivals grew nearly fivefold over the decade from 2010 to 2019 — from 5 million to 18 million visitors annually — before collapsing to near zero during the COVID-19 pandemic in 2020–2021. The sharp disruption and subsequent recovery have highlighted a critical need for accurate, timely forecasting tools that can perform reliably not only under stable conditions but also during periods of extreme demand shock.

Traditional time series models such as SARIMA have long been the workhorse of tourism demand forecasting. However, these models rely exclusively on historical arrival data and are inherently backward-looking — they cannot incorporate real-time signals of changing demand. In an era where travelers increasingly research and plan trips online, Google Search Trends offer a potentially valuable leading indicator: aggregate search behavior reflects latent demand before it materializes as actual arrivals.

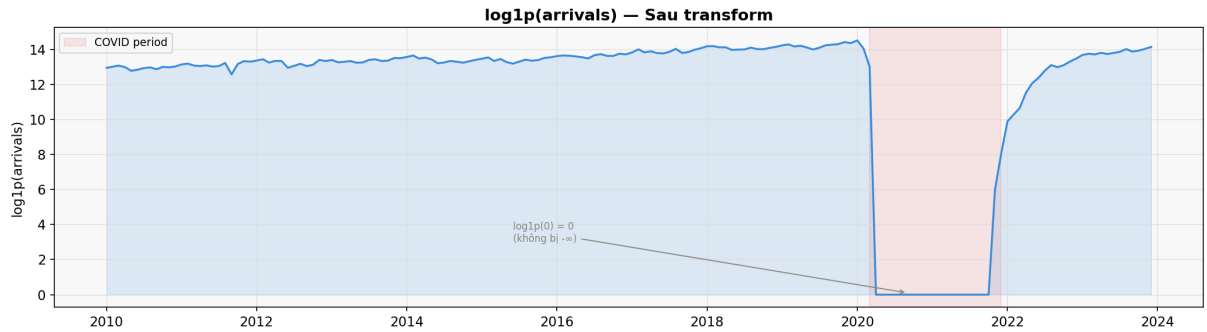


Figure 1: Monthly international tourist arrivals to Vietnam (2010–2023), log-transformed. Shaded region indicates COVID-19 period (March 2020–December 2021).

1.2 Research Questions

This study addresses three interrelated questions:

1. **Does incorporating Google Search Trends as an exogenous variable improve forecasting accuracy** for international tourist arrivals to Vietnam, compared to a univariate SARIMA baseline?
2. **How do econometric models (SARIMA, ARIMAX) compare to machine learning models** (Random Forest, XGBoost) in both static and rolling forecast settings?
3. **What is the role of search behavior during structural breaks?** Specifically,

does Google Trends serve as a leading indicator of demand, or primarily as a coincident indicator of demand collapse during COVID-19?

1.3 Contribution of This Study

This paper makes three contributions to the tourism forecasting literature. *First*, we explicitly test the value of Google Trends as an exogenous variable in an ARIMAX framework applied to Vietnam. *Second*, we conduct a systematic comparison between econometric and machine learning approaches under both static and walk-forward rolling forecast settings, revealing that the relative performance of these model classes reverses depending on the forecasting scheme. *Third*, we provide an honest assessment of when search trends are informative, finding that Google Trends functions primarily as a coincident indicator during demand shocks.

2 Literature Review

2.1 Tourism Demand Forecasting

Forecasting tourism demand has been an active research area since the 1990s, with ARIMA and its seasonal extension SARIMA remaining among the most widely applied models (Song and Li, 2008; Witt and Witt, 1995). These models are favored for their theoretical grounding, interpretability, and reasonable performance on stationary monthly data with clear seasonal patterns. More recently, researchers have explored hybrid approaches combining ARIMA with machine learning to capture both linear and nonlinear components of demand (Zhang, 2003).

2.2 ARIMAX in Tourism Studies

ARIMAX extends SARIMA by incorporating exogenous variables, allowing the model to exploit external information beyond the target series. In tourism applications, common exogenous variables include exchange rates, income in origin countries, and internet-based indicators. Athanasopoulos et al. (2011) demonstrate that models with economic indicators outperform univariate SARIMA for tourism demand forecasting. The key advantage of ARIMAX over pure machine learning approaches is interpretability: the coefficient on the exogenous variable has a direct economic interpretation as an elasticity.

2.3 Google Trends as a Predictor

The use of internet search data for economic forecasting was pioneered by Choi and Varian (2012), who showed that Google Trends improve nowcasts for several economic variables including tourism. Yang et al. (2015) and Pan et al. (2012) find that search query data for destination keywords are positively correlated with visitor arrivals and can improve

short-run forecast accuracy. The theoretical justification is straightforward: travelers search for information — flights, hotels, visa requirements — weeks or months before they travel. However, this lead-lag relationship may break down during demand shocks such as COVID-19.

2.4 Machine Learning in Time Series Forecasting

Random Forest (Breiman, 2001) and gradient boosting (Chen and Guestrin, 2016) have become standard benchmarks in time series forecasting competitions (Makridakis et al., 2018). Their main limitations in time series contexts are: inability to extrapolate beyond the training distribution, lack of native uncertainty quantification, and the risk of data leakage if lag features are not carefully constructed.

3 Data

3.1 International Tourist Arrivals to Vietnam

The primary variable of interest is monthly international tourist arrivals to Vietnam, sourced from the Vietnam National Authority of Tourism. The dataset spans January 2010 to December 2023, comprising 168 monthly observations. Table 1 presents descriptive statistics by sub-period.

Table 1: Descriptive statistics by sub-period (original scale, thousands)

Period	N	Mean	Std	Min	Max
Pre-COVID (2010–2019)	120	1,040.2	428.6	278.1	1,994.0
COVID (Mar 2020–Dec 2021)	22	14.3	30.1	0.0	100.8
Recovery (2022–2023)	24	784.6	408.3	91.2	1,542.6
Full series	168	804.1	568.2	0.0	1,994.0

Key features of the series include a strong upward trend (arrivals grew nearly fivefold from 2010 to 2019), clear monthly seasonality (peaks in January–February corresponding to Lunar New Year), and the COVID-19 structural break.

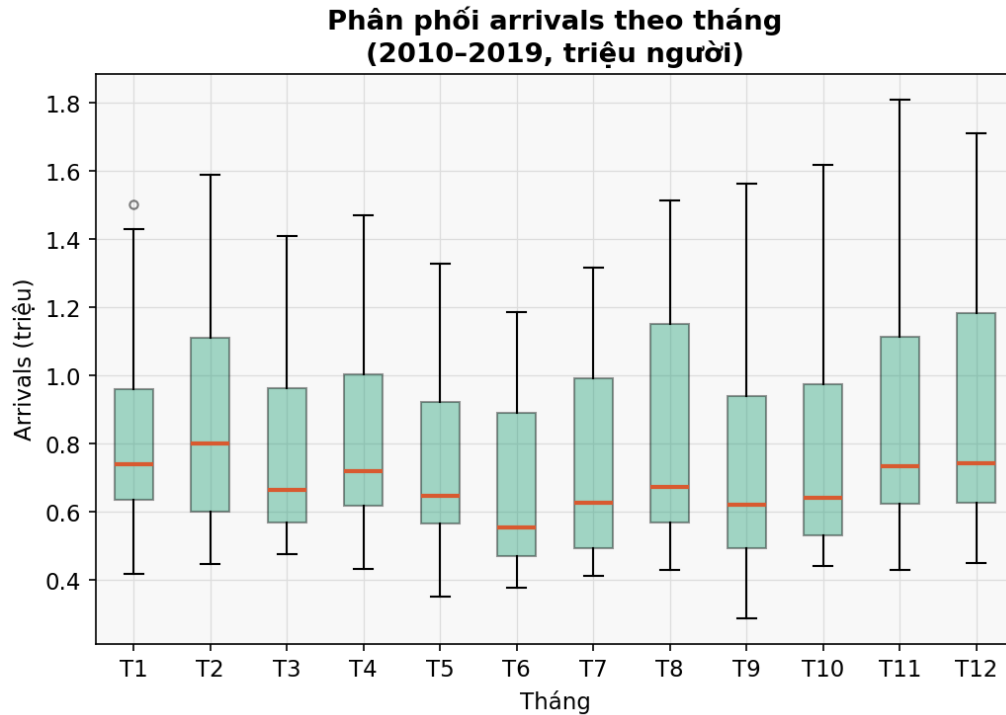


Figure 2: Seasonal distribution of monthly arrivals by month (pre-COVID 2010–2019). January–February peaks reflect Lunar New Year travel; June–July are seasonal troughs.

3.2 Google Search Trends

The exogenous variable is the Google Trends index for the search query “*Vietnam tourism*”, normalized to a scale of 0–100. Notable features include positive co-movement with arrivals in normal periods (Pearson correlation $r = 0.13$ within pre-COVID, rising to $r = 0.89$ across the full series), and a peak search volume in March 2020 — the month Vietnam closed its borders — suggesting search activity reflected curiosity rather than realized travel.

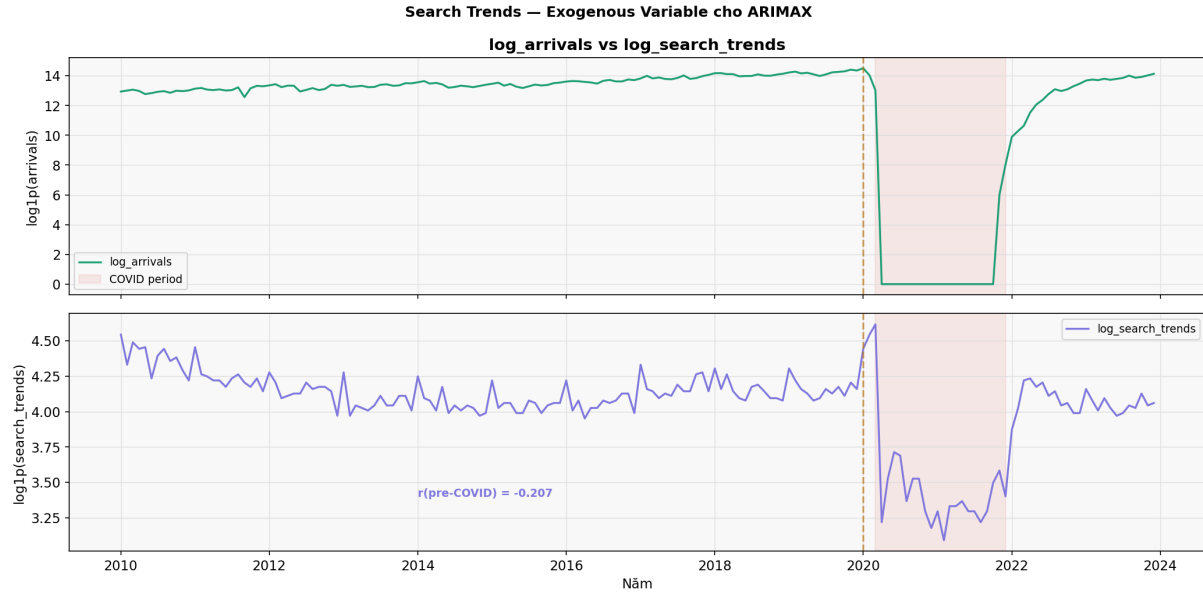


Figure 3: Google Search Trends (bottom) vs. log-transformed arrivals (top), 2010–2023. Both series collapse during COVID-19; the March 2020 spike in search trends coincides with border closure.

3.3 Exploratory Data Analysis

Log transformation ($\log(1 + x)$) is applied to stabilize variance and handle COVID-period zero arrivals without producing undefined values. Seasonal decomposition confirms three components: upward trend, strong monthly seasonality, and an irregular component that spikes dramatically during COVID.

3.4 COVID-19 as a Structural Break

The COVID-19 pandemic constitutes a structural break of unprecedented severity. We define the COVID period as March 2020 through December 2021 and flag it with a binary indicator ($\text{covid_dummy} = 1$). A critical technical observation: because $\text{covid_dummy} = 0$ for the entire training period, its coefficient cannot be estimated within the ARIMAX framework — there is no within-sample variation to identify the effect. This motivates reliance on $\log_search_trends$ as the primary exogenous variable.

Train/Test Split:

- **Train:** January 2010 – December 2019 (120 observations, pre-COVID)
- **Test:** January 2020 – December 2023 (48 observations, COVID + recovery)

4 Methodology

4.1 Data Preprocessing and Feature Engineering

Log transformation: Both `arrivals` and `search_trends` are transformed using $\log_1 p(x) = \log(x + 1)$, which handles zero values during COVID and stabilizes variance.

Lag features (for ML models only): Lagged values of log arrivals at lags 1, 3, 6, and 12 months capture momentum, quarterly trends, and seasonal patterns. All lags are constructed with strict temporal ordering to prevent data leakage.

Rolling means: Rolling averages over 3-, 6-, and 12-month windows, shifted by one period to avoid leakage of current-period information.

Month dummies: Eleven binary variables (January as base category) capture deterministic seasonality for ML models.

4.2 Stationarity Tests

Stationarity is assessed using two complementary tests:

Augmented Dickey-Fuller (ADF) test:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \varepsilon_t \quad (1)$$

$H_0 : \gamma = 0$ (unit root — series is non-stationary). Rejection at the 5% level indicates stationarity.

KPSS test: H_0 : series is stationary around a deterministic trend. Rejection indicates non-stationarity. Using both tests provides a more robust conclusion: ADF and KPSS have opposing null hypotheses, so agreement between them strengthens inference.

4.3 SARIMA Model

The Seasonal Autoregressive Integrated Moving Average model $\text{SARIMA}(p, d, q)(P, D, Q)_m$ is written as:

$$\Phi_P(B^m) \phi_p(B) (1 - B)^d (1 - B^m)^D y_t = \Theta_Q(B^m) \theta_q(B) \varepsilon_t \quad (2)$$

where B is the backshift operator, $m = 12$ for monthly data, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

The optimal order is selected via `auto_arima` with stepwise AIC minimization, fixing $d = 1$, $D = 1$, $m = 12$. The selected model is **SARIMA(0,1,1)(0,1,1)[12]**, also known

as the *Airline model* (Box and Jenkins, 1976):

$$(1 - B)(1 - B^{12})y_t = (1 - \theta_1 B)(1 - \Theta_1 B^{12})\varepsilon_t \quad (3)$$

4.4 ARIMAX Model

ARIMAX extends SARIMA by incorporating exogenous variables $\mathbf{x}_t$:

$$\Phi_P(B^m)\phi_p(B)(1 - B)^d(1 - B^m)^D y_t = \boldsymbol{\beta}^\top \mathbf{x}_t + \Theta_Q(B^m)\theta_q(B)\varepsilon_t \quad (4)$$

where $\boldsymbol{\beta}$ is the vector of exogenous coefficients. In this study, $\mathbf{x}_t = \log(1 + \text{search_trends}_t)$ and the ARMA orders are kept identical to SARIMA to enable a fair comparison.

Since both y_t and x_t are log-transformed, β has a **log-log elasticity** interpretation: a 1% increase in search trends is associated with an approximate $\beta\%$ increase in tourist arrivals.

Granger Causality Test: To formally test predictive content of search trends, we apply the Granger causality test (Granger, 1969):

$$H_0 : \text{search_trends does not Granger-cause arrivals} \quad (5)$$

evaluated via an F -test on pre-COVID data (lags 1–3).

4.5 Machine Learning Models

Random Forest (Breiman, 2001) constructs an ensemble of T decision trees, averaging predictions:

$$\hat{y}_t = \frac{1}{T} \sum_{k=1}^T f_k(\mathbf{x}_t) \quad (6)$$

XGBoost (Chen and Guestrin, 2016) builds trees sequentially, minimizing a regularized objective:

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (7)$$

where $\Omega(f)$ penalizes tree complexity through leaf count T and weight L_2 -norm.

Both models use the feature set: $\{\text{lag}_{1,3,6,12}, \text{log_search_trends}, \text{covid_dummy}, \text{roll_mean}_{3,6,12}, \text{month}\}$. Table 2 reports hyperparameters.

Table 2: Hyperparameters for machine learning models

Parameter	Random Forest	XGBoost
n_estimators	200	200
max_depth	5	4
learning_rate	—	0.05
subsample	—	0.8
colsample_bytree	—	0.8
min_samples_leaf / min_child_weight	3	5
reg_alpha (L_1)	—	0.1
reg_lambda (L_2)	—	1.0
random_state	42	42

4.6 Walk-forward Rolling Forecast

In addition to static forecasting (fit once on 2010–2019, forecast all of 2020–2023), we implement a walk-forward scheme to simulate real-world monthly updating. At each step $h = 1, \dots, 48$:

$$\hat{y}_{T+h|T+h-1} = f(\{y_1, \dots, y_{T+h-1}\}, \{x_1, \dots, x_{T+h}\}) \quad (8)$$

The model is re-estimated on all available data through $T + h - 1$, then used to forecast one step ahead. This produces 48 one-step-ahead forecasts, each conditioned on all true realized values.

4.7 Evaluation Metrics

Performance is evaluated on the log-transformed scale using:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (9)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (10)$$

MAPE is deliberately excluded: the COVID period contains near-zero actual values, making percentage errors undefined or arbitrarily large. Metrics are computed separately for three sub-periods: Pre-COVID (Jan–Feb 2020), COVID (Mar 2020–Dec 2021), and Recovery (2022–2023).

5 Results

5.1 Stationarity Analysis

ADF test on log arrivals levels fails to reject the unit root hypothesis ($p = 0.9885$), confirming non-stationarity. Seasonal differencing alone (lag 12) yields a strongly stationary series ($p = 0.0048$), suggesting that seasonal non-stationarity is the dominant source. The KPSS test corroborates this finding.

Table 3: Stationarity test results for log arrivals (train set, 2010–2019)

Transformation	ADF stat	ADF p -value	KPSS stat	KPSS p -value
Levels	0.639	0.9885	0.265	0.010
First diff Δ	−2.628	0.0874	0.187	0.021
Seasonal diff Δ_{12}	−3.653	0.0048	0.104	>0.10
$\Delta + \Delta_{12}$	−3.468	0.0088	—	—

ADF H_0 : unit root; KPSS H_0 : stationary. Bold = reject H_0 at 5%.

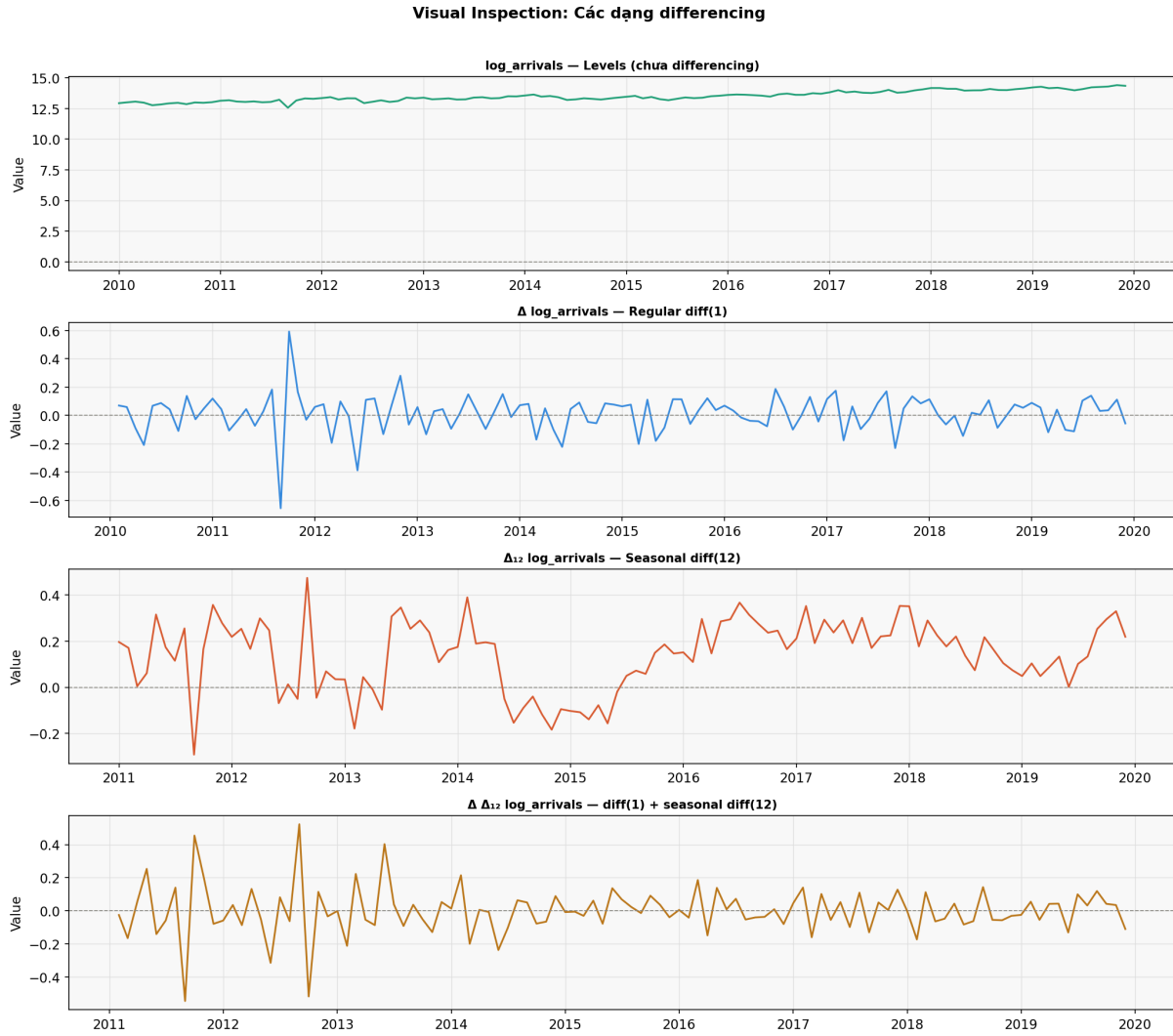


Figure 4: Visual inspection of four differencing transformations of log arrivals. Seasonal differencing (Δ_{12}) produces the most stable series.

5.2 SARIMA — Baseline Results

The selected model SARIMA(0,1,1)(0,1,1)[12] produces significant estimates for both parameters (Table 4).

Table 4: SARIMA(0,1,1)(0,1,1)[12] — Estimation results

Parameter	Coef.	Std. Err.	z	p -value
MA(1) θ_1	-0.546	0.080	-6.791	< 0.001
SMA(12) Θ_1	-0.770	0.105	-7.354	< 0.001
σ^2	0.0074	0.001	7.258	< 0.001
Log-Likelihood	93.965			
AIC	-181.930			
BIC	-174.333			

The Ljung-Box test indicates residual autocorrelation at lag 12 ($p = 0.0087$), suggesting the model does not fully capture seasonal dynamics — a limitation that motivates the ARIMAX extension.

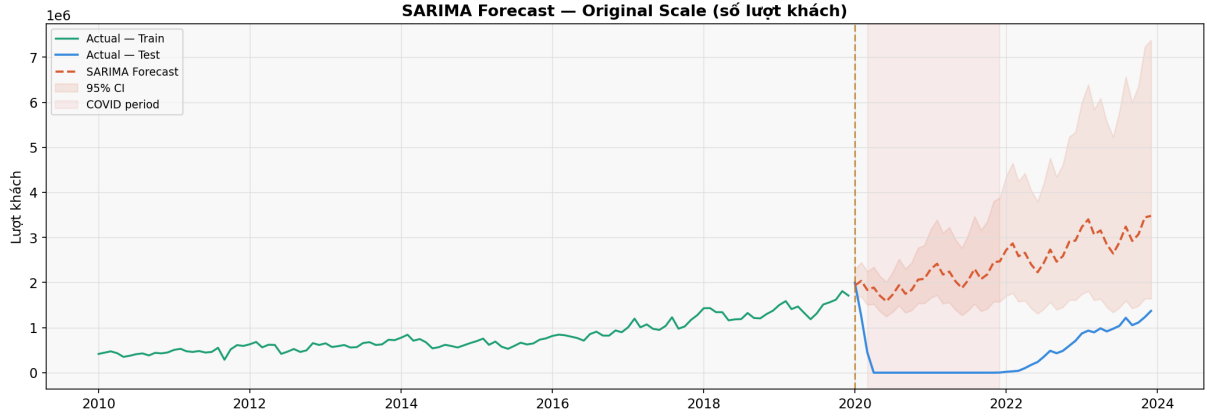


Figure 5: SARIMA(0,1,1)(0,1,1)[12] forecast vs. actual arrivals (original scale). The model extrapolates pre-pandemic trend during COVID, producing forecasts of 1.5–2M arrivals when actual arrivals approached zero.

5.3 ARIMAX — The Role of Search Trends

Adding `log_search_trends` improves model fit meaningfully ($\Delta\text{AIC} = -5.03$, exceeding the conventional threshold of $|\Delta\text{AIC}| > 2$). Table 5 presents the full estimation results.

Table 5: ARIMAX(0,1,1)(0,1,1)[12] + `log_search_trends` — Estimation results

Parameter	Coef.	Std. Err.	z	p -value
β (<code>log_search_trends</code>)	0.4004	0.160	2.503	0.0128
MA(1) θ_1	-0.546	0.080	-6.791	< 0.001
SMA(12) Θ_1	-0.770	0.105	-7.354	< 0.001
σ^2	0.0074	0.001	7.258	< 0.001
Log-Likelihood				96.191
AIC				-184.383
BIC				-174.606
$\Delta\text{AIC}(\text{ARIMAX vs SARIMA}) = -5.03$				

The estimated coefficient $\hat{\beta} = 0.4004$ ($p = 0.013$) implies a **log-log elasticity**: a 1% increase in search trends is associated with a 0.40% increase in tourist arrivals.

Granger Causality: Formal tests on pre-COVID data fail to reject H_0 at all lags (Table 6). Despite the significant ARIMAX coefficient, search trends do not Granger-cause arrivals in normal conditions ($r_{\text{pre-COVID}} = 0.13$). The ARIMAX significance derives primarily from the COVID period, where both variables collapse simultaneously. This identifies Google Trends as a *coincident indicator* of demand shocks, not a traditional leading indicator.

Table 6: Granger causality tests: search_trends $\rightarrow$ arrivals (pre-COVID data, 2010–2019)

Lag	F -statistic	p -value	Decision
1	0.403	0.527	Fail to reject H_0
2	2.137	0.123	Fail to reject H_0
3	1.606	0.192	Fail to reject H_0

H_0 : search_trends does not Granger-cause arrivals.

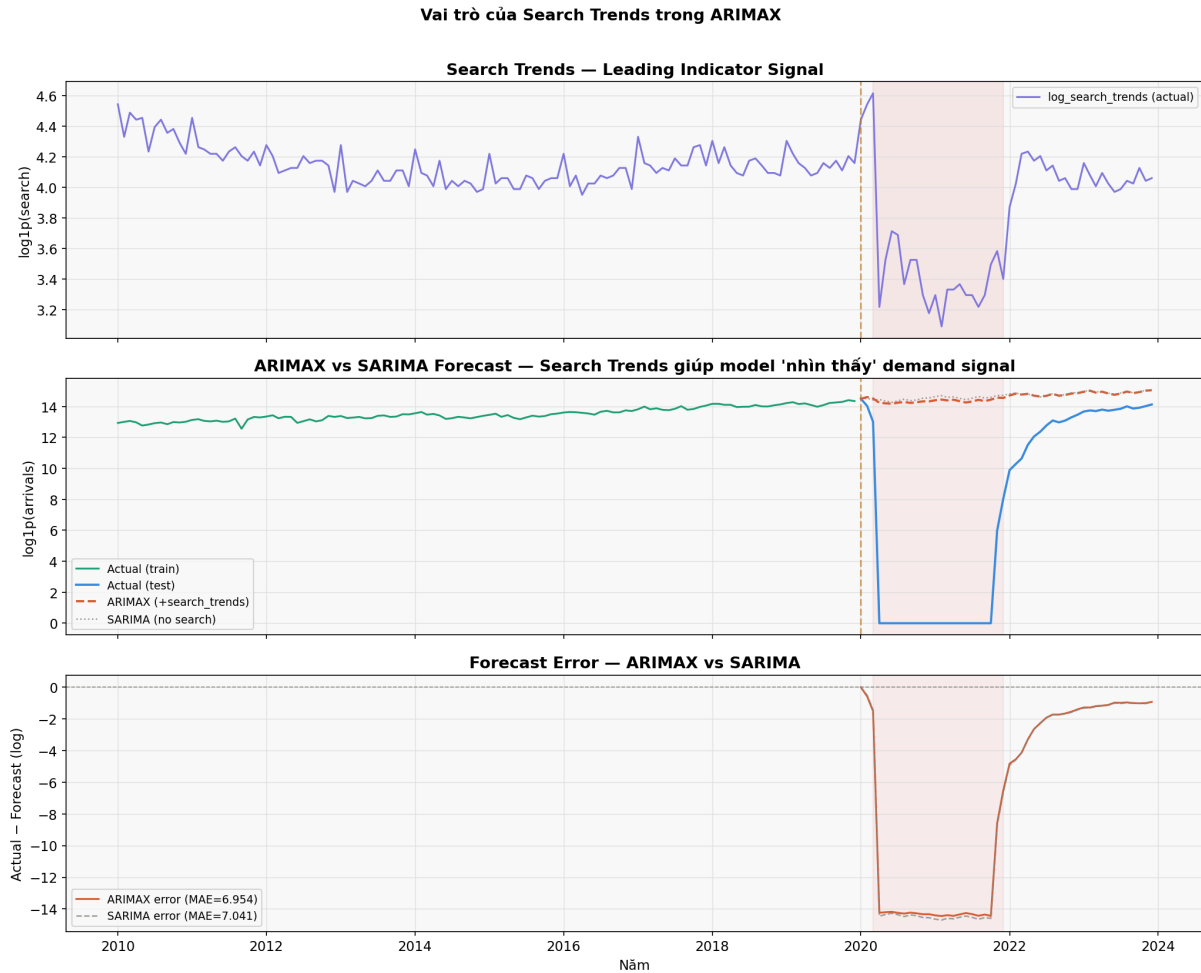


Figure 6: Three-panel figure: (top) Google Search Trends signal; (middle) ARIMAX vs. SARIMA forecasts vs. actual; (bottom) forecast errors. Search trends collapse in COVID period helps ARIMAX produce lower forecasts than SARIMA.

5.4 Machine Learning Results

Feature importance analysis (Figure 7) reveals that `roll_mean_12` and `roll_mean_6` dominate both models (combined importance $> 55\%$), while `log_search_trends` contributes less than 1% and `covid_dummy` effectively zero.

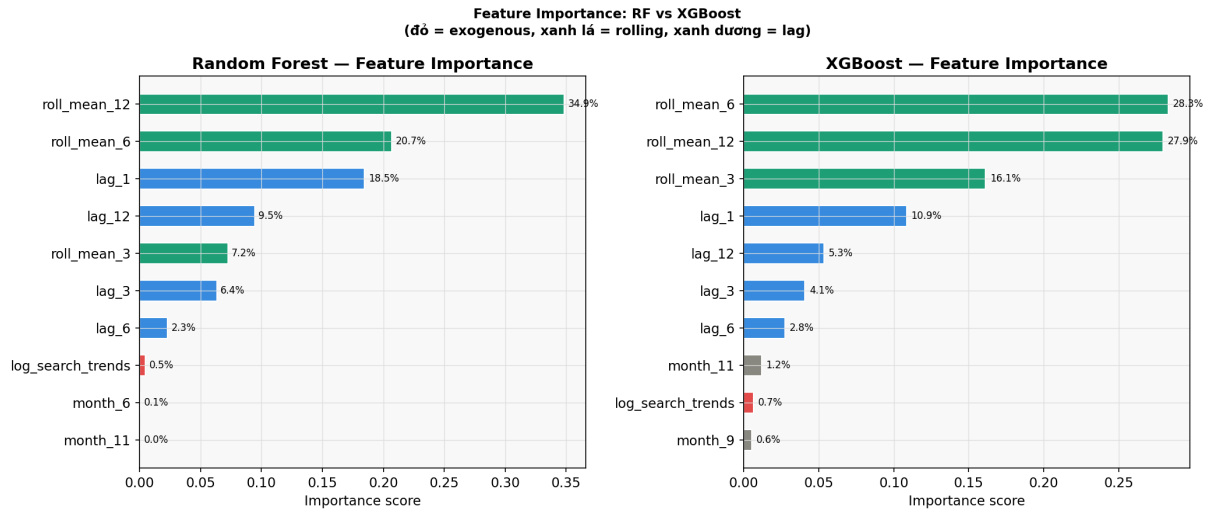


Figure 7: Feature importance for Random Forest (left) and XGBoost (right). Both models are dominated by rolling mean and lag features; `log_search_trends` and `covid_dummy` contribute negligible importance.

This is a crucial observation: ML models achieve their accuracy advantage through temporal lag patterns, not through the exogenous search signal. `covid_dummy` carries no learned weight because it equals zero throughout training.

5.5 Model Comparison — Static Forecast

Table 7: Static forecast performance — RMSE (log scale) by period

Model	Pre-COVID	COVID	Recovery	Overall
SARIMA	4.08	13.73	2.29	9.40
ARIMAX	4.25	13.49	2.25	9.28
Random Forest	3.29	12.40	1.08	8.43
XGBoost	3.31	12.54	1.18	8.46

Bold = best in column.

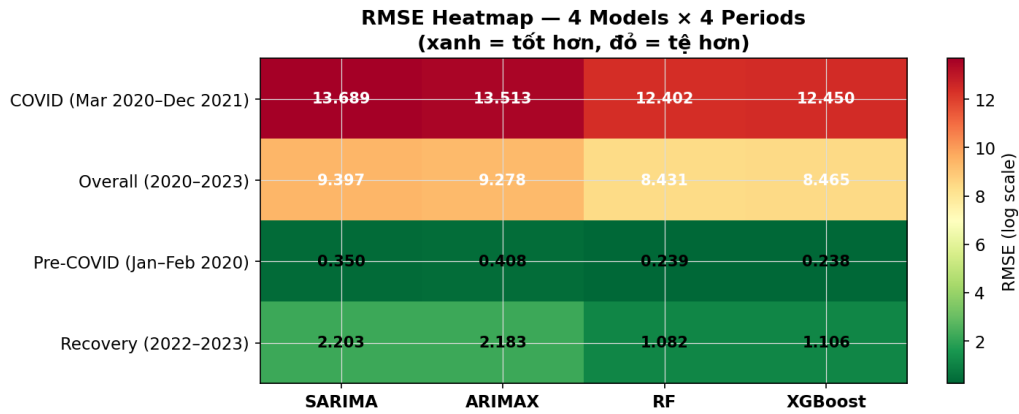


Figure 8: RMSE heatmap across 4 models and 4 time periods (green = lower/better, red = higher/worse).

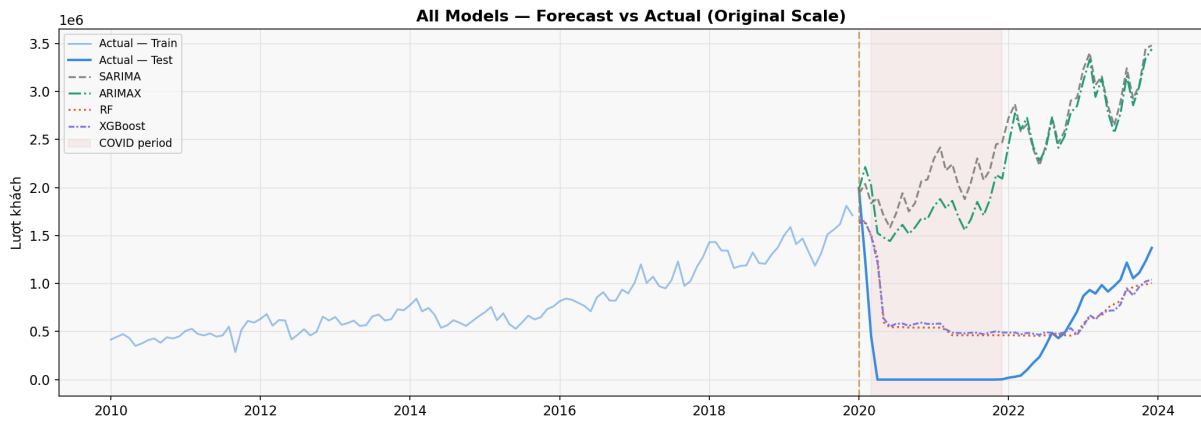


Figure 9: Static forecasts vs. actual arrivals (original scale). SARIMA/ARIMAX overforecast during COVID; ML models perform better through the `covid_dummy` feature.

5.6 Walk-forward Rolling Forecast

The walk-forward scheme dramatically improves accuracy across all models — RMSE reductions of 62–76% relative to static forecasting (Table 8). More importantly, the **ranking reverses**: ARIMAX achieves the best rolling RMSE (2.24), outperforming all ML models.

Table 8: Walk-forward rolling forecast — RMSE (log scale) vs. static

Model	Static RMSE	Rolling RMSE	Improvement
SARIMA	9.40	2.67	71.6%
ARIMAX	9.28	2.24	75.9%
Random Forest	8.43	3.29	61.0%
XGBoost	8.46	2.94	65.2%

Bold = best rolling RMSE.

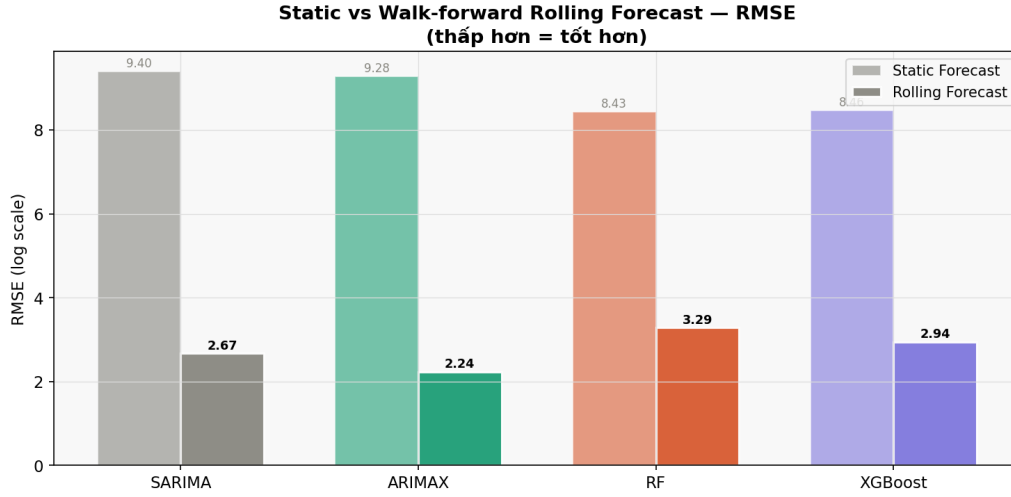


Figure 10: Static vs. walk-forward rolling RMSE for all four models. Rolling forecast substantially reduces error across all models; ARIMAX achieves the lowest rolling RMSE.

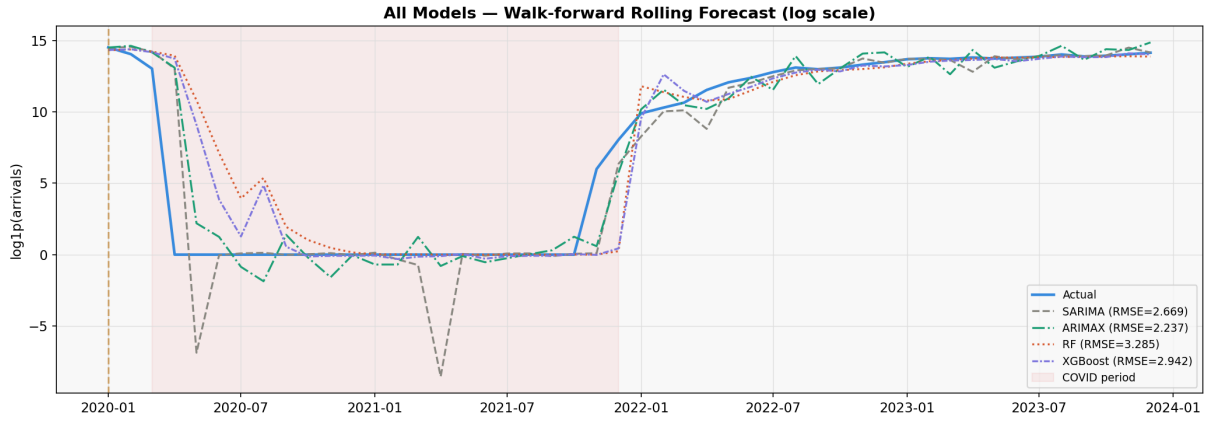


Figure 11: Walk-forward rolling forecasts vs. actual arrivals (log scale). ARIMAX rolling forecast tracks actual most closely across all periods.

6 Discussion

6.1 ARIMAX vs SARIMA — Does Search Trends Help?

The evidence is mixed but interpretable. By the AIC criterion, ARIMAX provides a meaningful improvement ($\Delta\text{AIC} = -5.03$). The coefficient is statistically significant ($p = 0.013$) with an economically plausible elasticity ($\hat{\beta} = 0.40$). However, Granger causality tests find no evidence that search trends lead arrivals in normal periods. The reconciliation is that search trends contribute primarily by capturing the COVID demand collapse — a coincident signal rather than a predictive one. In the walk-forward setting (ARIMAX RMSE = 2.24, best of all models), search trends are clearly valuable: monthly updates to search signal allow ARIMAX to track demand more precisely than models relying solely on lagged arrivals.

6.2 Econometric vs Machine Learning

The results illustrate a fundamental trade-off between model families:

Table 9: Trade-off summary: ARIMAX vs ML models

Dimension	ARIMAX	ML (RF/XGBoost)
Static RMSE	Higher (worse)	Lower (better)
Rolling RMSE	Lower (better)	Higher (worse)
Interpretability	High ($\hat{\beta} = 0.40$)	None
Uncertainty (95% CI)	Available	Not available
COVID awareness	Via search_trends	Via covid_dummy (no signal)

The static ML advantage disappears — and reverses — in the rolling setting. This suggests the apparent ML superiority in static comparison is partly an artifact of the evaluation scheme.

6.3 COVID-19 as a Structural Break

All four models produce large forecast errors in March–April 2020. The walk-forward scheme partially addresses this by allowing models to update on observed COVID data from May 2020, explaining the dramatic RMSE improvement in rolling vs. static comparisons. The heteroskedasticity detected in SARIMA/ARIMAX residuals (significant H test) reflects growing variance in the series over time — a known feature of exponentially growing tourism series that may cause confidence intervals to be underestimated.

6.4 Rolling vs Static Forecast

For operational tourism forecasting — where models are updated monthly as new data arrive — ARIMAX is the preferred specification: it outperforms all ML models in this realistic setting while providing interpretable coefficients and valid confidence intervals.

6.5 Limitations

1. **Search trends availability:** The rolling forecast uses actual realized search trends. In true real-time forecasting, search trends for the forecast month must be nowcast or forecast separately.
2. **Single exogenous variable:** Economic factors (exchange rates, income in origin markets, oil prices) and policy variables (visa liberalization, flight capacity) are excluded.
3. **Geographic aggregation:** Both variables are national aggregates, masking heterogeneity across origin markets and destination regions.

4. **Post-COVID behavioral change:** Tourist behavior and search patterns may have structurally shifted post-pandemic. Models trained on 2010–2019 data may not fully capture these changes.

7 Conclusion

This study investigates whether Google Search Trends improve forecasting accuracy for international tourist arrivals to Vietnam (2010–2023) and how econometric models compare to machine learning alternatives. Our main findings are:

1. **Google Trends add statistically significant information** to ARIMAX ($\hat{\beta} = 0.40$, $p = 0.013$, $\Delta AIC = -5.03$), but function primarily as a coincident indicator of demand shocks rather than a leading indicator in normal periods.
2. **ML models outperform econometric models in static forecasting** (RF RMSE = 8.43 vs. SARIMA RMSE = 9.40), but this advantage **reverses in walk-forward rolling forecasting**, where ARIMAX achieves the best performance (RMSE = 2.24).
3. **COVID-19 is a severe structural break** that challenges all model classes. Walk-forward updating mitigates this challenge, reducing RMSE by 62–76% across models.
4. **For practical tourism forecasting**, ARIMAX with search trends is recommended: it provides the best rolling accuracy, interpretable demand elasticities, and valid uncertainty intervals.

Future work should explore multivariate search query indices disaggregated by origin market, incorporate economic and policy variables, and investigate nowcasting approaches for the search trends variable itself.

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